

Homework:

- 1) Determine whether the following integral is convergent or divergent

$$\int_1^{\infty} \frac{\cos^2 x}{x^2+1} dx$$

- 2) Determine whether the following series are convergent or divergent

a) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$

b) $\sum_{n=3}^{\infty} \frac{5n^4 - 3n^3 + 2}{n^2 \cdot (n+1)(n^2+5)}$

- 3) Determine whether the following integral convergent or divergent

$$\int_1^{\infty} \frac{1 + \sin^2(e^{\cos x})}{\sqrt{x}} dx$$

- 4) a) Determine if the following series is convergent. If so, find the sum of the series.

$$\sum_{n=1}^{\infty} \frac{1}{4n^2-1}$$

b) $\int_0^{\pi/6} \frac{\cos x}{\sqrt{1-2\sin x}} dx = ?$

5) $\int \sqrt{1+\cos x} dx = ?$

6) $\int \frac{x^2}{\sqrt{(1+x^2)^3}} dx = ?$

7) $\sum_{n=1}^{\infty} [\arctan(n+1) - \arctan(n)]$ is given.

Find the sum of the given series

8) $\sum_{a=3}^{\infty} \frac{1}{a \cdot \ln a \sqrt{\ln(\ln a)}}$ is given.

Determine if it is convergent or divergent

9) $\sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right)$ is given.

Determine if it is convergent or divergent

Homework

A.1) $\int_1^{\infty} \frac{\cos x^2}{x^2+1} dx$ (Direct comparison test)

$\frac{|\cos x^2|}{x^2+1} \leq \frac{1}{x^2+1} < \frac{1}{x^2} > 0$ on $[1, \infty)$. Thus,
we can use the comparison test. For $\forall x \in [1, \infty)$, $\int_1^{\infty} \frac{1}{x^2} dx$ (p-integral, $p=2 > 1$),
so, it is convergent.

Thus, $\int_1^{\infty} \frac{\cos^2 x}{x^2+1} dx$ is also convergent.

A.2) a) $\sum_{n=1}^{\infty} \left(1 + \frac{1}{n}\right)^{-n^2}$

Let $a_n = \left(1 + \frac{1}{n}\right)^{-n^2}$ for $n \geq 1$. Then $a_n > 0$ for all n and

$\sqrt[n]{a_n} = \left(1 + \frac{1}{n}\right)^{-n}$. Hence,

$\lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{-n} = e^{-1} < 1$

By root test, we find that the series is convergent.

b) $\sum_{n=3}^{\infty} \frac{5n^4 - 3n^3 + 2}{n^2(n+1)(n^2+5)}$

Let $a_n = \frac{5n^4 - 3n^3 + 2}{n^2(n+1)(n^2+5)}$, $b_n = \frac{1}{n}$. Then,

$\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{5n^4 - 3n^3 + 2}{n^2(n+1)(n^2+5)} \cdot \frac{n}{1} = 5 > 0$

By p-test, we know that $\sum_{n=1}^{\infty} \frac{1}{n}$ is divergent.

Hence, by limit comparison test, the given series is divergent.

A.3) For $\forall x \geq 1$ $1 + \sin^2(e^{\cos x}) \geq 1$

Because, For $\forall x \geq 1$ $\sin^2(e^{\cos x}) = (\sin(e^{\cos x}))^2 \geq 0$

For $\forall x \geq 1$ $0 \leq \frac{1}{\sqrt{x}} \leq \frac{1 + \sin^2(e^{\cos x})}{\sqrt{x}}$

$\int_1^{\infty} \frac{1 + \sin^2(e^{\cos x})}{\sqrt{x}} dx$ is divergent because

$\int_1^{\infty} \frac{1}{\sqrt{x}} dx$ divergent. (Direct comparison test)

A.4) a) $\sum_{n=1}^{\infty} \frac{1}{4n^2-1} = \sum_{n=1}^{\infty} \frac{1}{2} \left(\frac{1}{2n-1} - \frac{1}{2n+1} \right)$

The partial sums of this series

$$S_k = \sum_{n=1}^k \frac{1}{(2n-1)(2n+1)} = \sum_{n=1}^k \frac{1}{2} \left[\frac{1}{2n-1} - \frac{1}{2n+1} \right]$$

$$= \frac{1}{2} \left(1 - \frac{1}{3} + \frac{1}{3} - \frac{1}{5} + \dots + \frac{1}{2k-1} - \frac{1}{2k+1} \right)$$

$$= \frac{1}{2} \left(1 - \frac{1}{2k+1} \right)$$

$\lim_{k \rightarrow \infty} S_k = \frac{1}{2}$ and the series converges.

The sum of the series is the limit of the partial sums.

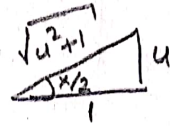
$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} = \frac{1}{2}$$

b) $\int_0^{\pi/6} \frac{\cos x}{\sqrt{1-2\sin x}} dx = \lim_{\epsilon \rightarrow 0} \int_0^{\frac{\pi}{6}-\epsilon} \frac{\cos x}{\sqrt{1-2\sin x}} dx$ $1-2\sin x = t$
 $2\cos x dx = dt$

$$\lim_{\epsilon \rightarrow 0} \frac{1}{2} \int_0^{\frac{\pi}{6}-\epsilon} \frac{dt}{\sqrt{t}} = \lim_{\epsilon \rightarrow 0} \sqrt{t} \Big|_0^{\frac{\pi}{6}-\epsilon} = \lim_{\epsilon \rightarrow 0} \left(\sqrt{1-2\sin(\frac{\pi}{6}-\epsilon)} - 1 \right) = -1$$

$$A.5) \int \sqrt{1+\cos x} \, dx = ?$$

$$u = \tan \frac{x}{2} \Rightarrow dx = \frac{2}{1+u^2} du$$



$$\cos \frac{x}{2} = \frac{1}{\sqrt{u^2+1}}$$

$$\int \sqrt{1+\cos x} \, dx = 2\sqrt{2} \int \frac{du}{(u^2+1)^{3/2}}$$

$$\cos x = 2 \cos^2 \frac{x}{2} - 1$$

$$1 + \cos x = 1 + 2 \cos^2 \frac{x}{2} - 1$$

$$1 + \cos x = 2 \cdot \left(\frac{1}{\sqrt{u^2+1}} \right)^2 = \frac{2}{u^2+1}$$

$$\sqrt{1+\cos x} = \sqrt{\frac{2}{u^2+1}}$$

$$u = \tan v \Rightarrow du = \sec^2 v \, dv$$

$$= 2\sqrt{2} \int \frac{\sec^2 v \, dv}{(\tan^2 v + 1)^{3/2}} = 2\sqrt{2} \int \frac{dv}{\sec v}$$

$$1 + \tan^2 v = \sec^2 v$$

$$= 2\sqrt{2} \int \cos v \, dv = 2\sqrt{2} \sin v$$

$$= 2\sqrt{2} \sin(\arctan u)$$

$$= 2\sqrt{2} \sin(\arctan(\tan \frac{x}{2}))$$

$$= 2\sqrt{2} \sin \frac{x}{2} + C$$

$$A.6) \int \frac{x^2}{\sqrt{(1+x^2)^3}} dx = ?$$

$$x = \tan t \Rightarrow dx = \sec^2 t dt$$

$$\boxed{\sec t = \sqrt{1+x^2}}$$

$$\int \frac{\tan^2 t \cdot \cancel{\sec^2 t}}{\cancel{\sec^3 t} \sec t} dt = \int \frac{\sec^2 t - 1}{\sec t} dt$$

$$= \int \sec t dt - \int \cos t dt$$

$$= \ln(\sec t + \tan t) - \sin t + C$$

$$= \ln(x + \sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$$

$$A.7) \sum_{n=1}^{\infty} [\arctan(n+1) - \arctan(n)]$$

$$S_k = \sum_{n=1}^k [\arctan(n+1) - \arctan(n)]$$

$$= \cancel{\arctan 2} - \cancel{\arctan 1} + \arctan 3 - \cancel{\arctan 2} + \dots + \underline{\arctan(k+1)} - \underline{\arctan k}$$

$$\lim_{k \rightarrow \infty} S_k = \lim_{k \rightarrow \infty} [\arctan(k+1) - \arctan 1]$$

$$= \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$$

$$A.8) \sum_{a=3}^{\infty} \frac{1}{a \cdot \ln a \cdot \sqrt{\ln(\ln a)}}$$

For $\forall n$; $u_{n+1} < u_n$. So, Apply the integral test

$$\int_3^{\infty} \frac{da}{a \ln a \sqrt{\ln(\ln a)}} = \lim_{b \rightarrow \infty} \int_3^b \frac{du}{u \cdot \sqrt{\ln u}} \quad \begin{array}{l} \ln u = t \\ \frac{1}{u} du = dt \end{array}$$

$$\begin{array}{l} \ln a = u \\ \frac{1}{a} da = du \end{array}$$

$$= \lim_{b \rightarrow \infty} \int_3^b \frac{dt}{\sqrt{t}}$$

$$= \lim_{b \rightarrow \infty} (2\sqrt{t}) \Big|_3^b$$

$$= \lim_{b \rightarrow \infty} (2\sqrt{\ln(\ln a)}) \Big|_3^b \rightarrow \infty$$

Divergent.

$$A.9) \sum_{n=1}^{\infty} \ln\left(1 + \frac{1}{n}\right) \rightarrow \text{(Apply integral test)}$$

It will be divergent